

3. (4 points) Find $\lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x^3}} \int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2) dt$.

4. (4 points) The solution between 3 and 4 of the equation $\tan x = 0$ is π . Starting with $x_1 = 3$, find the second approximation x_2 to π . (Use $\sin 6 = -0.30$)

5. Let $I_n = \int_0^1 (1-x^2)^n dx$ where n is a positive integer.

(a) (2 points) Find the number k such that $I_{20} = kI_{19}$.

(b) (3 points) Find I_{20} .

6. (4 points) Suppose the graph of the function $y = x^3 + px^2 + qx$ intersects the parabola $y = x^2$ when $x = 0$, $x = 1$ and $x = a$, where $a > 1$. If the two regions between the curves have the same area, find the volume of the solid obtained by rotating the two regions about the y -axis.

7. (3 points) Evaluate the indefinite integral:

$$\int \frac{1}{x\sqrt{3x^2 - 2x - 1}} dx. \text{ (Hint: Substitute } x = \frac{1}{t} \text{)}$$

8. (5 points) Evaluate the definite integral:

$$\int_0^{\pi} \left[\frac{\cosh x}{\cosh x + \cosh(\pi - x)} + \frac{x \sin x}{1 + \cos^2 x} \right] dx.$$

9. (5 points) Evaluate the indefinite integral:

$$\int \left(\frac{1}{\sin^2 x \cos^2 x} + \frac{\sec^2 x}{\tan^3 x - 1} \right) dx.$$